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Autonomous Wave Energy Vessel (AWEV)

Technical Report: Autonomous Wave Energy Vessel (AWEV)

Emission-free wave-powered cargo vessel — An innovation platform for the shipping of the future

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Summary

This report introduces the Autonomous Wave Energy Vessel (AWEV), a groundbreaking concept for emission-free global shipping.
The system harnesses the kinetic energy of sea waves as the primary power source, which is efficiently converted into electricity via turbines integrated in the vessel hull.
The generated energy propels the vessel forward and allows surplus storage in high-capacity batteries.
By eliminating the need for fossil fuels, AWEV is entirely energy-neutral and emission-free.
The goal is to catalyze a radical transition of the shipping industry by providing a self-sufficient transport system based on nature's own forces — developed without commercial self-interest, as an open research contribution.

1. Background and the Global Emissions Challenge

The shipping sector is a significant contributor to global climate emissions, currently responsible for approximately 3% of carbon dioxide emissions. Despite progress in biofuels and alternative fuels, the fundamental dependence on external energy supply remains.

AWEV addresses this challenge by presenting a genuine paradigm shift: a self-sufficient system that entirely eliminates fuel needs by converting ambient wave energy into propulsion and

stored energy.

2. Architecture and Energy Conversion

AWEV is designed as a hydrodynamic unit consisting of two integrated modules:

2.1 The Surface Vessel (Kinetic Energy Capture)

The hull is characterized by through-going, specially designed channels (resembling a Swiss-cheese structure for minimal friction) that allow sea waves to pass straight through.

A bidirectional turbine arrangement (horizontal and vertical) is placed in these channels to maximize capture of the wave motion energy regardless of direction.
The energy harvest is fine-tuned in real time through the use of modified Archimedes screws with adjustable flaps.

This construction ensures that the turbines dynamically optimize power extraction based on wave height and frequency, while keeping resistance to the vessel's forward motion to a minimum.

2.2 The Submerged Cargo Module (Stability and Payload Capacity)

The main cargo container is positioned approximately 10 meters below the water surface, a depth where wave influence and seaway are substantially reduced, providing an extremely stable transport environment.
The module is streamlined and suspended in vertical shafts.
It functions as active ballast, effectively countering roll and enhancing the vessel's seaworthiness.

3. Autonomous Cargo Flow Management

The cargo hold houses a sophisticated system for independent container handling.

AI-based sorting robots manage cargo transfer and stacking on roller-based transport tracks.

At port calls, the vessel's ballast is continuously and automatically maintained: one container is lowered through a shaft while another is lifted from the other shaft.
This balanced method secures uninterrupted stability and fast turnaround times.

4. Energy Strategy and Surplus Storage

AWEV operates according to a zero-fuel principle:
All propulsion energy is generated and recycled within the system solely through wave power.
Surplus energy is channeled into efficient battery units on the vessel's deck. Solar collectors on the upper deck serve as a marginal reserve for system operation during completely calm conditions.

5. Artificial Intelligence for Navigation and Logistics

AWEV is designed for unmanned operation and full autonomy, governed by an advanced AI system.

Adaptive Route Optimization:

The AI analyses wave height, direction and ocean data in real time to dynamically adjust the route.
This ensures maximum energy production rather than minimal distance.

Predictive Logistics:

The system computes transport times based on expected wave conditions and coordinates the logistics chain.
This enables slow, continuous and energy-efficient deliveries, eliminating the need for energy-wasting high-speed surges.

Global Range:

On wave-rich routes such as the North Atlantic, AWEV can maintain constant, emission-free propulsion year-round.

6. Key Advantages and Strategic Impact

AWEV offers significant advantages across several domains:

Environment / Climate:

Zero emissions of carbon dioxide, nitrogen oxides and particulates.
This provides a radical reduction of shipping's climate footprint.

Operational Economy:

Elimination of fuel costs.
Operational expenses are reduced to maintenance of the electromechanical and autonomous system.

Energy:

Completely self-sustaining.
The system utilizes a globally available and inexhaustible resource: the ocean's kinetic energy.

Hydrodynamics:

Exceptional stability thanks to the submerged cargo module, which reduces cargo damage and improves seaworthiness.

Logistics:

Predictive, AI-driven logistics enables optimized goods flow and reduces “just-in-time” stress.

7. Focus Areas for Further Technical Development

To realize AWEV, research and development is required in:

Advanced Turbine Design (LFAS)

Development of corrosion-resistant, robust and high-efficiency turbines and Archimedes screw/flap systems for long-term operation in marine environments (IAKKS)

Materials Science: development of lightweight composite materials with optimal tolerance to seawater and fatigue.

AI Integration: further research into adaptive route planning, integration with satellite-based weather systems and predictive maintenance analysis.

8. Societal Value and Open Source Philosophy

This concept represents not only a technical solution but a societal vision. AWEV eliminates fuel dependence and creates a scalable model for future shipping — from coastal freight to transoceanic systems.

Most importantly:

This design is intended as an open research contribution. No patents will be sought, as the aim is to remove economic barriers and accelerate global development toward sustainability.

9. Appeal to Global Research

I, Göran Skoog, have developed and presented this system for purely altruistic reasons, focusing on environmental benefit. I appeal to engineering schools, research institutes and technical universities to promptly initiate simulations and practical prototype developments based on this platform.

A realization of AWEV would constitute a historic breakthrough for truly emission-free and self-sustaining global shipping.

Lift-force-optimized Archimedes screw for wave energy conversion (LA-Screw)

Technical Report – Lift-force-optimized Archimedes screw for wave energy conversion

Appendix to: Autonomous Wave Energy Vessel (AWEV)

Author: Göran Skoog
Date: 2025-10-21

Summary

This report describes an advanced version of the Archimedes screw, adapted for installation in AWEV's internal wave channels.
The construction is based on the principle of creating lift in water similar to an airplane wing in air, which generates a continuous rotation moment when waves pass through the vessel.
The system utilizes both the waves' vertical energy and their forward motion to achieve higher efficiency and stable energy extraction without external moving components.

1. Introduction

The purpose of the improved Archimedes screw is to increase energy extraction from wave motions through a combination of pressure differentials and hydrodynamic lift.
Unlike a traditional screw that relies solely on flow movement along the spiral, this construction uses aero-hydrodynamic profiling of the screw blades.
Each spiral turn functions as an airfoil that creates lower pressure on the upper side when water flows past, which significantly amplifies rotational force.

2. Principle and Function

When a wave passes through the vessel's channel, alternating pressure changes set the water in motion.
The lift-force-optimized Archimedes screw is mounted in this channel and is designed with twisted airfoil profiles (twisted blades) that capture the wave flow and convert it into rotation.
The aerodynamic similarity to an airplane wing is central: the upper side of the spiral blade has a longer flow path, creating lower pressure, while the underside has a shorter flow path, leading to higher pressure. The difference generates a lift moment that enhances rotation.

3. Construction and Placement

The screw is completely enclosed in the vessel's canal system. No parts protrude from the hull, thereby e
liminating the risk of cavitation, wear or drag in open sea.
The channels are designed so that incoming wave power is directed through a series of converging flow profiles that focus the energy toward the active surfaces of the screw blades.
The internal placement also protects the system from mechanical impact and reduces noise and vibration.

4. Lift and Twist Optimization

The screw's efficiency is directly linked to how lift is generated along the spiral length.
By introducing a controlled twist — i.e., a gradual change of blade angle from center to periphery — lift can be kept nearly constant despite varying flow speeds in the wave profile.
With optimal twist and profiling, even small wave motions can generate significant rotational torque, rendering the construction efficient even under low sea activity.

5. Research and Development Areas

The following areas are judged central for further development:

1. Profile development – exploring different airfoil profiles (NACA-like curves) for maximum lift in a water environment.

2. Twist optimization – analysis of how variation in blade angle affects smoothness, efficiency and load distribution.

3. CFD simulation – computation of turbulence, lift distribution and cavitation at various wave frequencies.

4. Materials technology – investigation of lightweight composite materials with damping properties for long service life.

5. Integrated control – possibility for variable blade angle to adapt the system to different wave types and loads.

6. Expected Potential

By combining the Archimedes screw's mechanical simplicity with the wing principle's pressure-difference-driven lift, a hybrid concept is created that can yield significantly higher energy per wave unit than existing wave energy converters.
In the AWEV system, the screw simultaneously functions as an energy converter and a wave stabilizer, contributing to calmer motion in seaway.

7. Conclusion

The lift-force-optimized Archimedes screw represents a new way to exploit wave energy in closed maritime systems.
By letting waves pass through the vessel and using their natural motion to generate rotation via hydrodynamic lift, an effective and sustainable energy conversion is achieved.
Further research should focus on airfoil shape, twist optimization, flow dynamics and material composition to determine ideal geometry and efficiency.

Integrated, Active, Ceramic Composite Coating System (IAKKS) for Marine Energy Systems and Turbine Components

Technical Report: Integrated, Active, Ceramic Composite Coating System (IAKKS)

For Marine Energy Systems and Turbine Components

Author: Göran Skoog

Summary

This report presents a groundbreaking integrated coating system (IAKKS) designed to address the combined challenges of erosion, fatigue and biological fouling in extreme marine environments.
The system is directly applicable to ship bottoms, high-performance turbine screws, wave energy devices and offshore structures.

The system's novelty lies in its unique combination of three technologies:

Ceramic wear resistance inspired by brake-pad technology.

Active, non-toxic antifouling through an embedded electrical grid pulsing low voltage.

Structural reinforcement that prevents delamination and maintains surface smoothness over long time.

The goal is a service life of at least 20 years *without repainting* while maintaining maximum hydrodynamic efficiency on all critical components.

1. Purpose and Novelty

Traditional marine coatings cannot simultaneously offer wear resistance, long lifetime, surface smoothness and non-toxicity.

IAKKS solves this by integrating functions that were previously separate: mechanical protection, biological protection and structural strength.

The novelty is precisely the integration:

A system where ceramic hardness, electrical antifouling and reinforced composite work together in a single active material.

2. System Construction

2.1 Coating Components

Binder:

Two-component thixotropic epoxy- or polyurethane-based binder, constituting the wear-resistant coating matrix. Its high viscosity at rest ensures form stability on vertical surfaces.

Wear-resistant Particles:

Titanium dioxide (TiO_2) and ceramic grains of the same type used in brake pads, which provide high frictional stability, abrasion resistance and thermal tolerance to the coating.

Conductive Additive:

Graphite or carbon nanotubes (CNT) to provide electrical conductivity.

Reinforcement:

A corrosion-resistant conductive mesh in titanium or nickel-plated carbon fiber embedded in the mass.

2.2 Functional Mechanism

Mechanical abrasion resistance:
The ceramic particles protect against cavitation, sand, ice and pressure shocks.

Active antifouling:

A low-voltage pulse (1–10 Hz, 1–12 V) passes through the mesh and prevents cells from attaching.

Structural strength:

The reinforcement mechanically binds the layer to the substrate and prevents crack propagation.

Rheology (Form stability):

The coating's thixotropic properties (syrup-like at rest) ensure that it fills the mesh and retains its shape on vertical surfaces without sagging during curing.

3. Application

3.1 High-precision (turbines, screws)

Surface is roughened and cleaned.
The mesh is affixed to the component.
The composite mass is applied and vacuum-treated to eliminate air pockets (ensuring zero porosity).

Curing in controlled temperature.
Wet sanding and polishing to mirror finish (< 10 µm roughness).

3.2 Large-scale surfaces (hulls, channels)

Hull is blasted and the mesh attached in sections.
The coating is sprayed or rolled on.
Post-curing and polishing as appropriate.

4. Technical Advantages

< 20-30 years of service life:

Ceramic hardness and reinforcement protect against erosion and cracking.

No biological fouling:

Pulsed antifouling prevents attachment without environmental toxins.

Constant hydrodynamic surface:

Mirror finish maintains highest efficiency.

Environmental protection:

Completely non-toxic, safe for fish and marine ecosystems.

Resistance to ice and cavitation:

Brake-pad-inspired particle mixture withstands extreme stresses.

5. Research and Development Needs

Fatigue testing:

Investigate reinforcement's long-term effect under cyclic loading.

Electrical verification:

Measure mesh resistance changes after 10,000 hours of saltwater exposure.

Process development for high precision:

Develop and verify a CNC- or robot-controlled method for application and subsequent sanding/polishing.
The goal is to guarantee dynamic balance and hydrodynamically smooth surface (< 10 µm) on the sensitive turbine screw.

Biological testing:

Long-term testing in real marine environment (6–12 months).

Cavitation and freeze/thaw simulation:

Testing in cavitation tunnel and freeze/thaw cycles.

6. Integration in Existing Concepts

IAKKS is developed to be directly applicable to prior marine innovations, such as the Lift-force-optimized Archimedes screw for wave energy conversion; this component and its channel walls, where combined wear resistance and perfectly maintained surface smoothness are absolutely critical for the system's maximum energy efficiency.

AWEV — vessel hull and internal wave channels where zero-fouling is critical for energy efficiency.

The coating eliminates the need for cleaning, improves hydrodynamics and extends operational lifetime.

The system makes the whole concept virtually maintenance-free (Lift-force-optimized Archimedes screw and AWEV) over long periods and can be applied across marine environments.

7. Technical Analysis and Criteria

Scientific basis:

Built on synthesis of established composite and active antifouling technologies.

Technical realism:

System components are industrially available.

Novelty:

The design constitutes a unique integrated solution with three functional protective layers.

Projected durability:

The target is over 20 years of operation, dependent on material optimization and verification.

Synergy effect:

Expected to markedly increase performance of the Lift-force-optimized Archimedes screw and AWEV.

8. Conclusion

The Integrated, Active, Ceramic Composite Coating System (IAKKS) is a technically realistic and innovative solution to the three main problems in marine environments: erosion, fouling and material fatigue.

By combining brake-pad-inspired wear resistance, active electrical antifouling and reinforced composite structure, a life expectancy and surface smoothness are achieved that previously have not existed in marine technical coatings.

IAKKS naturally integrates with previous sustainability concepts/inventions such as the Lift-force-optimized Archimedes screw and AWEV, as a holistic solution protecting all surfaces — primarily the highly stressed turbine screw — against biological and mechanical degradation.

Thus the system creates an almost maintenance-free marine energy platform. The system is practically feasible, scientifically sustainable and unique in its functional combination.
It therefore constitutes a strong basis for further development, testing and academic dissemination.

Appendix: Complementary Technical Clarification for IAKKS

This appendix aims to clarify in detail the system's electrical operational principle, particularly how absolute isolation against the vessel's metal hull is guaranteed while maintaining the antifouling effect at the outer surface.

1. Principles for Electrical and Galvanic Isolation

The IAKKS system is designed to eliminate the risk of galvanic corrosion through its internal structure and material composition.

Dielectric Encapsulation:

The embedded, conductive mesh (of corrosion-free material) is fully encapsulated in the non-conductive ceramic composite matrix (based on epoxy or polyurethane).
This matrix acts as a dielectric barrier.
Its primary task is to prevent any direct electrical contact (solid conduction) between the conductive mesh and the underlying metal hull.

Galvanic Protection Function:

This encapsulation is the primary protection against galvanic corrosion.
By electrically separating the mesh from the hull, no galvanic cell can form, ensuring that the active antifouling function does not transfer any negative corrosion potential to the vessel's metal structure.

2. Mechanism for Surface Activation (Pulsing Effect)

Despite the internal isolation, the active pulse reaches the seawater via field transfer at the interface.

Field Induction:

The low-voltage pulse (1–12 V) fed into the mesh creates an electric field.
This field is induced through the thin insulating composite matrix up to its outer boundary.

Ionic Contact:

At the interface between the composite coating and the seawater (the electrolyte), the field creates an instantaneous potential shift.
This voltage is sufficient to disturb the microbiological process needed for single-celled organisms to attach.

Localization:

The effect is strictly localized to the microscopic interface layer.
The weak voltage rapidly decays into the surrounding water, ensuring the pulsing is harmless to marine ecosystems and does not affect other vessels.

3. Material Requirements and Scalability

To achieve long life and high performance, high demands are placed on material purity and installation robustness.

Material Purity (Wear Resistance):

The ceramic wear-resistance inspired by brake-pad technology must use pure, non-toxic ceramic particles (e.g., **aluminium oxide**) embedded in the matrix.
This excludes using unprocessed waste or components containing toxic substances (such as copper), to strictly meet environmental requirements.

Zoned Integration:

For large-scale application on a vessel hull, it is necessary that the embedded mesh is installed in independent sections/zones.

This facilitates installation, streamlines maintenance and enables isolation and troubleshooting of individual zones in case of damage without taking the whole system out of operation.

Hydrodynamic Surface:

The material's hardness and fineness must be optimized to allow polishing to an extremely low surface roughness, which is crucial for maintaining LFAS turbine and AWEV vessel maximum

hydrodynamic efficiency.
Polishing requirements vary considerably depending on the application.

Dynamically Adaptive Turbine System (DALAS)

Technical Report – Dynamically Adaptive Turbine System (DALAS)

Extraction of Uni-directional Impulse Energy in the AWEV platform

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Date: 2025-10-24

1. Introduction and System Context

This report describes an adaptive development of the lift-force-optimized Archimedes screw (LFAS) developed by the author, referred to as the Dynamically Adaptive Lift-force-optimized Archimedes System (DALAS).

The basic LFAS turbine design is based on extracting energy from the oscillating flow that normally occurs when waves pass through the AWEV vessel's channels.

The development described here focuses on handling the special case where the vessel is exposed to a strong, short mechanical impulse (**slam or impact)** from a large wave.

The purpose of DALAS is to eliminate the harmful stopping effect that occurs at an impact and at the same time convert the potentially destructive energy into maximal uni-directional electrical extraction.

2. Principles for Dynamic Impulse Handling (DALAS)

The LFAS turbine is designed to act as a turbine driven by hydrodynamic lift (airfoil profiles in spiral geometry).

In DALAS mode, the flow handling is modified by integrating the LFAS- turbine with a linear movement mechanism.

2.1 Mechanism for Impulse Handling

The LFAS turbine is mounted on low-friction rails inside the wave channel, coupled to a Linear Permanent Magnet Generator (LPG).

When a slam (impact) occurs, the DALAS system allows the turbine to accelerate linearly in the same direction as the wave pulse.

This achieves two things:

Elimination of the Stopping Effect:

Instead of absorbing the impulse as a mechanical shock against the mounts, it is converted into controlled linear kinetic energy.

Flow Modification to a Uni-directional Pulse:

The linear movement (extraction) extends the time the turbine is exposed to the rapid flow.
The system thus dynamically shifts from handling oscillating flow to optimizing extraction from a uni-directional pulse during the short impact period.

2.2 Dual and Regenerative Extraction

In the adaptive impulse mode, energy is extracted from two sources simultaneously, making extraction exceptionally efficient during the short pulse:

Rotation:

The lift-force-optimized LFAS screw maintains high rotational speed during the extended exposure period, which is maximized.

Linear Motion:

The linear movement itself is converted into electricity via the LPG.

After the pulse has passed, the LPG is used for electrical regenerative recovery.

The generated energy drives the LPG as a motor to gently and energy-efficiently pull the turbine back to its starting position for the next wave.

3. Novelty and Level of Innovation

The DALAS concept represents a new hybridization within marine energy conversion:

LFAS Geometry:

The combination of the lift-force principle and the long, spiral-shaped Archimedes surface is unique and not found in other known turbine designs.

Active Impact Conversion:

The mechanical solution for converting a potentially damaging shock load into extractable linear energy by allowing the rotating turbine to move is new.

Existing systems damp the shock, but DALAS extracts energy from it.

Dual Extraction:

The system is the first known to integrate and extract electricity from both the rotating lift force and the linear impulse motion via regenerative recovery.

4. Analysis: Advantages, Disadvantages and Research Needs

4.1 Advantages

Elimination of Shocks:

Extends system lifetime and reduces maintenance costs by avoiding mechanical fatigue in bearings and mounts.

High Impulse Efficiency:

Dynamic adaptation maximizes energy extraction from the most energy-rich moments (the pulses) by prolonged exposure.

Regenerative System:

Electrical recovery minimizes energy losses compared to pure mechanical spring- or damping systems.

4.2 Disadvantages and Challenges

Increased Mechanical Complexity:

The moving mechanism (rails, seals, LPG mounts) increases the number of critical components compared to a static turbine, which can raise maintenance requirements.

Control System Dependency:

The system's efficiency is entirely dependent on a fast and correct adaptive control system that can predict the wave pulse in real time and adjust the LPG's resistance.

Friction and Wear:

Movement in a marine environment increases the risk of corrosion and friction in the rails.

4.3 Research Needs

Regenerative Control:

Development of advanced algorithms for active dynamic control that optimize linear extraction and recovery to maximize net energy.

Fluid-Structure Interaction (CFD):

Detailed simulations of how the LFAS turbine interacts with linear movement in a turbulent, uni-directional pulse environment.

LPG Integration:

Research into the optimal design of the LPG to balance electricity generation from linear motion and its role as a recovery motor.

5. Conclusion

The DALAS concept is an innovative system solution addressing a known bottleneck in wave energy conversion:
handling high impulse energy.

The system is technically groundbreaking in its hybridization of lift-driven rotation and regenerative linear motion.

The aim of this report is to present the concept for scientific validation and further development within research projects and engineering studies.

Wall-integrated Shaftless Drive (RDG) and Passive Hydrostatic Stabilization (PHST)

TECHNICAL REPORT

Systemic Innovation for Marine Energy Extraction

Lift-force-optimized Rotor (LA-Screw) with Wall-integrated Shaftless Drive (RDG) and Passive Hydrostatic Stabilization (PHST)

Author: Göran Skoog
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Date: 2025-10-28

Summary

This report presents a holistic solution for efficient energy extraction in marine channel systems (tidal and wave power).

The combination of three innovative subsystems — the Lift-force-optimized Rotor (LA-Screw), the Wall-integrated Shaftless Drive (RDG) and the Passive Hydrostatic Stabilization (PHST) — constitutes an integrated concept for high efficiency, minimal mechanical loss and long lifetime in marine environments.

The LA-Screw contributes hydrodynamic efficiency, RDG eliminates gearbox and hub, and PHST solves the most critical mechanical limitation: **maintaining a microscopic air gap b**etween rotor and stator under dynamic load.

1. Background and Objectives

The Lift-force-optimized Rotor (LA-Screw) builds on a new hydrodynamic principle where energy is extracted via lift and a twisted airfoil profile rather than traditional mass displacement.
To fully realize its potential, a drive system without mechanical losses, vibrations and cavitation is required — something conventional shaft- and gearbox-systems cannot offer.

The goal is to create a mechanically simple, hydrodynamically pure and passively stabilized system that eliminates all unnecessary moving parts while guaranteeing maximum electrical and mechanical efficiency.

2. Wall-integrated Shaftless Generator (RDG)

The Wall-integrated Shaftless Generator (RDG) is the mechanical core of the system.

The generator's stator is integrated directly into the channel wall, while the rotating ring (rotor) is recessed and completely flush with the channel's inner surface.

This solution:

eliminates gearbox and central hub,

reduces turbulence and flow resistance,

and enables a completely sealed and maintenance-free enclosure.

Placement balances high flow exposure against protection from cavitation and impact loads.
The result is a compact, reliable and energy-efficient construction with potential for 3–10% higher system efficiency compared with conventional turbine systems.

3. The Critical Limitation

The decisive challenge in all wall-integrated, shaftless generators is maintaining a very small and stable air gap between rotor and stator during rotation.

The turbine axis and its surrounding components are always subject to radial wobble and vibrations from water pressure and flow variations. Conventional bearings accept this movement, which makes a minimal air gap impossible and limits efficiency.

It is this difficulty that PHST solves — without electronics, sensors or active control.

4. Passive Hydrostatic Stabilization (PHST)

PHST is a system for passive hydraulic position compensation that replaces sliding or rolling bearings with water-filled chambers.

The system maintains perfect concentric position between rotor and stator during rotation, despite shaft wobble.

PHST consists of concentric, water-filled components in stainless steel that rotate as a unit.

Main components (see accompanying sketch in original file):

Turbine shaft (A) — generates drive and unwanted wobble.

Primary isolation chambers (B) — counter shaft wobble through fluid-coupled chambers.

Inner positioning ring (C) — kept centered through hydraulic counterforce.

Stabilizing struts (D) — transfer corrected position to the rotor ring.

Rotor ring (E) — carries the generator magnets and is maintained in perfect position.

Fixed reference ring (F) — forms the zero point for the entire system.

Secondary fine-control chambers (G) — provide micro-correction against the reference ring.

5. Two-stage Hydraulic Compensation

Step 1 — Coarse isolation:

When the shaft wobbles, fluid is compressed in opposing chambers.

The pressure difference creates an immediate counterforce that stabilizes the inner ring and eliminates wobble.

Step 2 — Fine tuning:

Pressure differences between the secondary chambers restore the ring precisely to the centerline against the stationary reference ring.

This double feedback functions as a mechanical *equivalent to an active magnetic bearing, but is completely passive and self-regulating.*

6. Interaction between RDG and PHST

PHST is the component that enables practical realization of the RDG concept.

Without passive hydrostatic stabilization, it would not have been possible to maintain the microscopic air gap required for high electrical efficiency at low rotational speeds.

The integrated solution delivers:

Elimination of friction and mechanical tolerances.

Higher electrical efficiency through constant air gap.

No need for active control or electronics.

Reduced maintenance needs and higher operational reliability.

7. Research and Development

To confirm the system's potential the following is required:

CFD simulations to validate PHST's response time and hydrodynamic stability.

FEM analyses to evaluate RDG's structural stiffness and deformations.

Electromagnetic modelling to optimize efficiency at low rotational speeds.

Verification of sealing techniques and cavitation tolerance.

8. Conclusion

The proposed holistic solution — LA-Screw + RDG + PHST — represents a systemic novelty within marine energy extraction.

Together they form a self-stabilizing, shaftless and high-efficiency system that unites hydrodynamic efficiency, mechanical simplicity and passive precision.

This is a solution addressing all known limitations in current marine turbine systems and enabling energy extraction with higher stability, lower maintenance and maximal electrical efficiency — entirely without active control.

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